

Caddy Image Gallery

Operator Documentation



A Caddy v2 module for serving image and video galleries with a unified lightbox, configurable sort/pagination, and a templatable UI.

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Configuration

Caddyfile directive

```
1 handle_path /images/* {  
2     root * /var/www/html/images  
3     image_gallery  
4     file_server  
5 }
```

The `image_gallery` directive accepts one inline option:

Subdirective	Value	Default	Purpose
template	file name, relative to the templates dir	gallery.tmpl	Pick which template file to render. Path-traversal protected: no .., no absolute paths — the templates dir is a chroot.
no_thumbs	true / false (no-arg = true)	false (thumbs on)	Skip on-the-fly WebP thumbnail generation for images. Tile <code></code> points to the original file instead of <code>~/_thumbs/<name>.webp</code> . Thumb requests fall through to the next handler. Useful for small galleries where you don't want a thumb cache. See <code>no_thumbs</code> walkthrough below.

Subdirective	Value	Default	Purpose
no_video_thumbs	true / false (no-arg = true)	false (video thumbs on, if ffmpeg available)	Skip on-the-fly WebP thumbnail generation for videos (extracted from the first frame via ffmpeg). When true, videos still display in the gallery (with the placeholder gradient + play button on each tile) but no per-frame thumbnail is generated. When false (default), video thumbs ARE generated IF ffmpeg is available on the host. If ffmpeg is missing, video thumbs fall back to the placeholder regardless of this setting (we can't decode a frame without a tool that can). Use no_video_thumbs to skip the ffmpeg invocation even when it's available (e.g., on hosts where you don't want the CPU cost of frame extraction). See "Video thumbnails (ffmpeg)" below.

Subdirective	Value	Default	Purpose
page_size	integer >= 1	50	How many image entries to show per page. Must be a positive integer; page_size 0 is rejected (use no directive, or set the explicit value you want). The pagination nav only renders when total pages > 1, so a 30-image gallery at the default 50 shows all 30 on a single page with no nav.

Example with a themed subdir:

```

1 handle_path /images/* {
2     root * /var/www/html/images
3     image_gallery {
4         template themes/dark/gallery.tpl
5     }
6     file_server
7 }
```

This loads `$GALLERY_TEMPLATES_DIR/themes/dark/gallery.tpl` (or falls back to the bundled template if the file doesn't exist on disk). The path is validated at Provision — an invalid name (e.g. `../etc/passwd`) fails Caddy startup, not at first request.

Example: skip thumbnail generation

```

1 handle_path /images/* {
2     root * /var/www/html/images
3     image_gallery {
4         no_thumbs
5     }
6     file_server
7 }
```

With `no_thumbs`: - Each tile's `<img src>` is the original image file (`./photo.jpg`), not `~/_thumbs/photo.webp` - No thumb generation, no cache, no CPU cost on first request - The browser downloads the full image per tile (bigger page payload, slower on dirs of large photos)

Use `no_thumbs false` to turn it back on (the default is off, so the directive is opt-in).

Example: disable video thumbnails

Video thumbnails (per the table above) require `ffmpeg`. If `ffmpeg` isn't available, the gallery falls back to the placeholder gradient + play button automatically — you don't need to do

anything. If `ffmpeg` IS available and you want to skip frame extraction (e.g., on a low-CPU host or for very large videos), use `no_video_thumbs`:

```
1 image_gallery {
2     no_video_thumbs
3 }
```

With this, video tiles show the placeholder gradient + play button (no `<img>` for the frame). The same `no_video_thumbs false` form re-enables frame extraction.

The video thumb generation uses `ffmpeg -vframes 1` to extract the first frame, scaled to fit the configured `thumb_width × thumb_height` (defaults 320×320). The output is a WebP, written to the same cache dir as image thumbs (`/var/cache/caddy-gallery` by default, override via `GALLERY_THUMB_CACHE_DIR`). Same caching rules as image thumbs (regenerate only when the source video's mtime is newer than the cache file).

If the operator has `ffmpeg` installed at a non-standard path, set `FFMPEG_PATH` env var (note: not yet implemented — current code uses `exec.LookPath("ffmpeg")` which only checks `$PATH`). All standard install paths (`/usr/bin/ffmpeg`, `/usr/local/bin/ffmpeg`, etc.) are picked up automatically. Best for: small galleries (< 100 images) where you don't want a thumb cache and the originals aren't huge. Not recommended for large galleries — the page payload goes from ~30 KB (with thumbs) to ~5 MB average (full images) for a 1,000-image dir.

Example: change the page size

```
1 handle_path /images/* {
2     root * /var/www/html/images
3     image_gallery {
4         page_size 100
5     }
6     file_server
7 }
```

This shows 100 image entries per page instead of the default 50. Tradeoffs: larger pages mean fewer HTTP requests, but each request returns a bigger HTML payload (and the server uses more memory per render). The pagination nav at the bottom of the page only renders when total pages > 1, so if your gallery has 30 images and you set `page_size 100`, you get all 30 on one page with no nav. URL query override: append `?page=2` to the gallery URL to jump to a specific page. `?page_size=N` is NOT a query param — page size is set in the Caddyfile only (per-request override would let the user request arbitrarily large pages and could DOS the server).

All other configuration (the `root *` for the image directory, the `handle /` `handle_path` for the route, the auth wrapper) is via the surrounding Caddyfile block, not the `image_gallery` directive. The module reads the on-disk directory from Caddy's per-request `root` variable, or from the `Root` JSON field if set explicitly.

JSON config (advanced)

Equivalent JSON, if you're configuring Caddy via the admin API or a config file rather than a Caddyfile:

```
1 {
2     "handler": "image_gallery",
3     "root": "/var/www/html/images"
```

```
4 }
```

The `Root` field is optional — the module falls back to the per-request root set by the surrounding `handle / handle_path` block. Set it explicitly only if you need to override the request-time root.

Environment variables

The plugin reads one environment variable:

`GALLERY_TEMPLATES_DIR`

Where it's read in code: `render.go`, in two functions — `loadTemplate()` (at request time, to find the on-disk override) and `writeBundledTemplates()` (at Caddy startup, to write the bundled templates so operators can see them). Both do the same thing:

```
1 dir := os.Getenv("GALLERY_TEMPLATES_DIR")
2 if dir == "" {
3     dir = "/etc/caddy/gallery-templates"
4 }
```

Default: `/etc/caddy/gallery-templates`. Created on first Caddy startup by `writeBundledTemplates()` with mode `0755`, owned by whatever user the Caddy `systemd` service runs as. If the template file already exists, it's left alone (operator overrides are preserved across restarts).

The directory it points to: the absolute path you set. The plugin does not create parent directories beyond the templates dir itself — it just `MkdirAlls` the final directory. So if you point it at `/srv/gallery-templates`, that path needs to be writable by the Caddy process.

How to set it for the live Caddy (the canonical way — `systemd` starts the process with a clean environment, so your shell's `env` doesn't reach the service):

```
1 # One-off (persists in the manager's env, inherited by all units)
2 sudo systemctl set-environment GALLERY_TEMPLATES_DIR=/etc/caddy/gallery-templates
3 sudo systemctl restart caddy
4
5 # Or persistently in the caddy service unit
6 sudo systemctl edit caddy
7 # Add:
8 # [Service]
9 #   Environment="GALLERY_TEMPLATES_DIR=/etc/caddy/gallery-templates"
10 # Or use EnvironmentFile= pointing at a file with the line
11 sudo systemctl daemon-reload
12 sudo systemctl restart caddy
```

How to set it for dev (`go test`, `xcaddy build`, or running a custom Caddy from your shell):

```
1 export GALLERY_TEMPLATES_DIR=/path
2 # then run your commands
```

Note: the test suite sets `GALLERY_TEMPLATES_DIR` to a temp dir via `TestMain`, so `go test` is isolated from your shell's value. The module always reads the `env` at request time, so changes to the `env` var take effect on the next Caddy restart — no rebuild needed.

What happens if the dir is unwritable or doesn't exist: `writeBundledTemplates()` logs a warning to stderr and continues. The bundled templates still serve fine (the on-disk file is a convenience, not a requirement). The next request falls back to the bundled constant.

There are no other env vars. Sort order, pagination, page size, and the thumb cache directory are all configurable in code (constants in the source) rather than via env vars.

In-code constants (not configurable)

These are baked into the source. They can be changed by editing the code, rebuilding, and restarting Caddy — but they're not exposed as Caddyfile directives or env vars. Listed here so you know what the defaults are and where to find them.

Constant	Value	Where
Thumbnail size	320px	thumbnails.go
Thumb Cache-Control max-age	86400 (24h)	thumbnails.go
Thumb format	WebP (lossless, via nativewebp)	thumbnails.go
Scan cache TTL	1 minute	gallery.go (<code>NewScanCache(time.Minute)</code>)
Thumb width (px)	320	thumbnails.go (the <code>resize</code>)

Note: “Thumbnail size” and “Thumb width (px)” both reference the same constant (320) — the maximum dimension of the generated thumbnail. If you wanted them to be different values (say, max-dim 320 and width 280 for a non-square crop), let me know and I'll split them into two separate constants.

Why these aren't configurable yet: most operators never need to change them. The thumb size and Cache-Control max-age are sensible defaults that work for the vast majority of galleries. The scan cache TTL is short enough (1 minute) that new files appear quickly without manual cache busting, and long enough that hot directories don't re-scan on every request. If you find yourself wanting to change one of these, that's a signal that the constant should probably be promoted to a config option — file an issue and we can discuss.

What `image_gallery` does NOT do

- It does not generate thumbnails at build time — thumbnails are generated on-the-fly on first request and cached in `/var/cache/caddy-gallery/`. See the wiki page for cache details.
- It does not handle uploads or modifications. It's read-only.
- It does not support nested directory pagination — when you click into a subdirectory, that subdirectory has its own gallery with its own pagination, but the parent dir's image list isn't merged in.
- It does not redirect from `/foo` to `/foo/`. If you request `/images/generated` (no trailing slash) and it's a directory, the module returns a 301 with a relative `Location: generated/` header (so the browser resolves it relative to the current URL). This is a workaround for Caddy's `handle_path` rewriting both `r.URL.Path` and `r.RequestURI`, which makes absolute reconstruction impossible inside the handler.

caddy_image_gallery — Complete configuration reference

A single-page reference for every configuration knob the `image_gallery` Caddy handler exposes. Use this as the “what can I configure” reference; for how to use individual features, see the per-topic docs linked below.

For per-topic deep dives: - `configuration.md` — Caddyfile directive details + env vars - `templates.md` — the template file + variables - `sort-and-pagination.md` — the URL query API

1. Caddyfile subdirectives (the primary way)

Inside an `image_gallery { ... }` block:

Subdirective	Value	Default	Purpose
<code>sort</code>	<code>mtime</code> / name (also accepts date as an alias for <code>mtime</code> — see Sort & Pagination to Aliases)	<code>mtime</code> (newest first)	Default sort field. Overridable per-request via <code>?sort=</code> .
<code>template</code>	file name, relative to the <code>templates</code> dir	<code>gallery.tmpl</code>	Which template file to render. Path-traversal protected.
<code>no_thumbs</code>	<code>no-arg = true</code> / explicit <code>false = false</code>	<code>false</code> (thumbs on)	Skip thumbnail generation. Tile <code></code> points to the original file.
<code>page_size</code>	integer <code>>= 1</code>	50	Image entries per page. Nav only renders when total pages <code>> 1</code> .
<code>thumb_width</code>	integer <code>>= 1</code>	320	Max width in pixels. Source is fit-within-bounds.
<code>thumb_height</code>	integer <code>>= 1</code>	320	Max height in pixels. Source is fit-within-bounds.
<code>thumb_format</code>	<code>jpeg</code> / <code>jpg</code> / <code>png</code> / <code>webp</code>	<code>webp</code> (lossless)	Output format. <code>jpeg</code> quality 75, <code>png</code> lossless, <code>webp</code> lossless.
<code>cache_scan</code>	integer <code>>= 1</code>	1	Scan cache TTL in minutes.
<code>thumb_ttl</code>	integer <code>>= 1</code>	1440	HTTP Cache-Control: <code>max-age</code> for thumbs, in minutes (= 24h default).

Full Caddyfile example (every directive set):

```
1 | handle_path /images/* {
```

```

2   root * /var/www/html/images
3   image_gallery {
4       sort name
5       template themes/dark/gallery.tmpl
6       no_thumbs false
7       page_size 100
8       thumb_width 480
9       thumb_height 320
10      thumb_format jpeg
11      cache_scan 5
12      thumb_ttl 60
13  }
14  file_server
15 }

```

2. JSON config fields (programmatic)

Same fields, json tags. Use caddy adapt or any JSON config producer to set them:

```

1  {
2    "root": "/var/www/html/images",
3    "sort": "name",
4    "template": "themes/dark/gallery.tmpl",
5    "no_thumbs": false,
6    "page_size": 100,
7    "thumb_width": 480,
8    "thumb_height": 320,
9    "thumb_format": "jpeg",
10   "cache_scan": 5,
11   "thumb_ttl": 60
12 }

```

cache is runtime-only (excluded from JSON via json:"-").

3. Per-request URL query parameters (no config needed)

Param	Values	Default	Effect
sort	mtime / name / type / size (also accepts date as an alias for mtime — see Sort & Pagination to Aliases)	inherits from Caddyfile	Sort field
order	asc / desc	depends on sort	Sort direction
page	integer >= 1	1	Which page (only meaningful when page_size causes pagination)

Deliberately NOT query-overridable: - ?page_size=N — would let users request arbitrarily

large pages and could DOS the server - `?thumb_format=N` — would let users force the server to recompute thumbs in any format - `?thumb_width=N` — same

The page size, format, and thumb dimensions are set in the Caddyfile only.

4. Environment variables

Var	Default	Purpose
GALLERY_TEMPLATES_DIR	/etc/caddy/gallery-templates	Where the module looks for (and writes) the on-disk template override. The only user-facing env var.

A second env var exists in the code for testing only: `GALLERY_THUMB_CACHE_DIR` (default `/var/cache/caddy-gallery`) — not documented as a user-facing knob.

See `configuration.md` to Environment variables for the full setup story (systemd unit, dev workflow, failure mode).

5. Caddyfile handler-level wiring (external, not `image_gallery`-specific)

These live in the surrounding Caddyfile, not inside the `image_gallery { ... }` block:

Caddyfile construct	Purpose	Example
<code>handle_path</code> (or <code>route</code>)	Strips URL prefix and routes to <code>image_gallery</code>	<code>handle_path /images/* { ... }</code>
<code>root *</code>	Per-request “root” var, read by <code>image_gallery</code> to find the directory to scan	<code>root * /var/www/html/images</code>
<code>file_server</code>	Fallthrough for direct file requests and 404s (must come AFTER <code>image_gallery</code>)	<code>file_server</code>
Auth (Authelia, <code>basic_auth</code> , etc.)	Wraps the route in auth	(auth) snippet (or <code>basicauth { ... }</code> block)

6. External: the Caddyfile as a whole

The live site typically looks like:

```
1 hermes.synapticloop.com {
2   import auth
3   route /images/* {
4     root * /var/www/html/images
5     image_gallery
6     file_server
7   }
8 }
```

The (auth) snippet is defined elsewhere in the Caddyfile and wraps every route in Authelia forward_auth at 127.0.0.1:9091.

7. Behavior modifiers (not configurable, hardcoded)

These are baked into the source. They could be promoted to config options if a use case comes up, but currently aren't:

Constant	Value	Where
Thumb JPEG quality (when thumb_format jpeg)	75	thumbnails.go
Video thumbs	not generated (videos show play-button overlay in the template)	thumbnails.go (source extensions are image-only)
Lightbox	page-scoped (50 images on the current page, not all 1,197)	render.go (lightbox JS in the template)
Other-files strip on a subdir page	"Other files" rendered as horizontal chips above the image grid	render.go (splitFiles)
Directories strip on every page	Always rendered, alphabetical, ignores sort	render.go (splitFiles + dirs sort)

8. Quick reference — what's where

If you're wondering "where is this knob?":

Want to configure...	Use
Sort field (default + per-request)	sort directive + ?sort= query
Pagination	page_size directive + ?page= query
Thumbnail on/off	no_thumbs directive
Thumbnail size	thumb_width / thumb_height directives
Thumbnail format (jpeg/png/webp)	thumb_format directive
Thumbnail browser cache TTL	thumb_ttl directive
Directory scan cache TTL	cache_scan directive
Template file (theme)	template directive (e.g. themes/dark/gallery.tmpl)
Templates directory (where to find templates)	GALLERY_TEMPLATES_DIR env var
Thumbnails cache directory	GALLERY_THUMB_CACHE_DIR env var (testing only)
Auth (who can view)	Caddyfile basicauth or (auth) snippet (external)
Route (URL prefix)	Caddyfile handle_path or route (external)
Image directory	Caddyfile root * (external)

Templates

The gallery is rendered by an `html/template` (Go's standard template engine). The template is bundled in the binary as a Go string constant (`galleryTemplate` in `render.go`), and on each Caddy startup the module also writes it to disk at `/etc/caddy/gallery-templates/gallery.tpl` (or `$GALLERY_TEMPLATES_DIR` if set). The on-disk file exists so operators can read the template without opening the binary, and so they can override individual templates by editing the file in place.

How the template loading works

1	loadTemplate(name string) reads \$GALLERY_TEMPLATES_DIR
2	
3	└─ name = gallery.Template (from Caddyfile `template` directive)
4	default "gallery.tpl" if the directive is absent
5	
6	└─ sanitizeTemplateName(name):
7	reject absolute paths, reject "." (any traversal)
8	
9	└─ /etc/caddy/gallery-templates/<clean name> exists?
10	└─ YES to parse that file directly (CSS+JS are inside it)
11	(the on-disk file is the active template)
12	└─ NO to use the bundled galleryTemplate constant
13	
14	└─ return *template.Template ready to RenderPage

The `template` Caddyfile directive is the operator-facing knob that picks the template file. See [docs/configuration.md](#) for the directive syntax and the path-traversal protection details.

Note on `no_thumbs`: the template is unaffected by the `no_thumbs` Caddyfile directive. The template always uses `{{.ThumbURL}}` for the tile `<img src>`; the `no_thumbs` flag changes the *value* of that field (to the original file URL when true, to the thumb URL when false), not the field name or its usage. So a template that works with thumbs also works with `no_thumbs` (and vice versa), with no template changes needed.

The template is a single self-contained file. The HTML, CSS (inside `<style>`), and JS (inside `<script>`) all live in one Go string constant (`galleryTemplate` in `render.go`) and one on-disk file (`/etc/caddy/gallery-templates/gallery.tpl`). There is no sub-template loading — the previous Phase 16 design that split them into 3 files (`gallery.tpl` + `style.css` + `lightbox.js`) was collapsed in Phase 17 for easier editing.

The bundled constant is the source of truth at build time. The on-disk file is an *override* layer; if you delete it, the module falls back to the bundled version automatically on the next request.

What's in the single template file

One file, `gallery.tpl`, ~16.7 KB / 574 lines, containing:

Section	What's in it	Approx lines
<code><!DOCTYPE> ... <style></code>	Document head, including the full CSS inlined as a <code><style></code> block	~10 + 365 (CSS)

Section	What's in it	Approx lines
<body>, <main>, <header>	Title, counts, sort indicator, sort bar	~50
Directories section	{{if .Directories}} ... {{end}} — the dirs chip strip	~7
Other files section	{{if .OtherFiles}} ... {{end}} — the others chip strip	~7
Images section	Sort info, paginated grid, per-tile HTML	~80
Pagination	{{if gt .TotalPages 1}} ... {{end}} — prev/next links	~15
<script> ... </script>	The full JS inlined (lightbox, open-in-new-tab, sort indicator)	~100 (JS)

The CSS rules and the HTML they apply to are interleaved top-to-bottom in the file, so when you scroll through the template you see the structure (HTML), the styling (CSS), and the behavior (JS) in document order. Easier to hand-edit than three separate files.

Template variables

The template receives a `PageData` struct. All fields you can reference from `{{ ... }}` are listed below.

Top-level fields (on `PageData`)

Field	Type	What it's for
{{.Title}}	string	The page title (rendered in <title> and the <h1>). Currently the directory name.
{{.PathPrefix}}	string	Prefix for relative links to files in the same directory. Usually <code>"/"</code> .
{{.ThumbPrefix}}	string	Prefix for thumbnail URLs. Usually <code>"/_thumbs/"</code> .
{{.Directories}}	[]FileView	Subdirectory entries. Always rendered in full (not paginated), case-insensitive alphabetical.
{{.OtherFiles}}	[]FileView	Non-media files (HTML, txt, etc.). Always rendered in full (not paginated).
{{.Images}}	[]FileView	The image + video tiles for the current page only. Paginated, sorted per the user's <code>?sort=&order=</code> choice.
{{.Page}}	int	Current page number (1-based).
{{.PageSize}}	int	Images per page (constant; default 50).

Field	Type	What it's for
{{.TotalImages}}	int	Total images in the directory (across all pages).
{{.TotalPages}}	int	Total page count.
{{.HasPrev}}	bool	True if {{.Page}} > 1.
{{.HasNext}}	bool	True if {{.Page}} < {{.TotalPages}}.
{{.Sort}}	SortSpec	The current sort. See below.

{{.Sort}} (a `SortSpec` struct)

Field	Type	What it's for
{{.Sort.Field}}	string	One of "mtime", "name", "type", "size". (The ?sort= URL param. "mtime" is the default.)
{{.Sort.Order}}	string	"asc" or "desc". (The ?order= URL param.)

Per-entry fields (on `FileView`)

When you {{range .Images}} (or `.Directories` / `.OtherFiles`), each iteration gives you a `FileView` with these fields:

Field	Type	What it's for
{{.Name}}	string	The basename ("photo.jpg", "subdir", etc.). Truncated with ellipsis in the live template if too long for the tile.
{{.Href}}	string	Relative link to the file. Use as <code></code> .
{{.ThumbURL}}	string	For images and videos, the relative thumbnail URL (e.g. <code>./_thumbs/photo.webp</code>). Empty string for non-media files — check {{if .ThumbURL}} before using.
{{.IsDir}}	bool	True for directories.
{{.IsImage}}	bool	True for image files.
{{.IsVideo}}	bool	True for video files. Videos go in the image grid with a play-button overlay (no <code></code> child); {{.ThumbURL}} is set but the live template doesn't render an <code></code> for them.
{{.IsOther}}	bool	True for non-media files (HTML, txt, etc.).

Field	Type	What it's for
{{.Type}}	string	Uppercase extension without the dot: "JPG", "DIR", "MP4", "HTML", etc.
{{.Size}}	string	Human-readable file size: "234 KB", "1.2 MB", etc.
{{.Date}}	string	Empty string for directories. ISO date "2006-01-02" (UTC-normalised). Empty string for directories.

Template functions (the funcmap)

The template engine has a few helper functions registered:

Func	Signature	What it does
minus1	minus1 n int to int	Returns $n - 1$. Used for prev-page link targets.
plus1	plus1 n int to int	Returns $n + 1$. Used for next-page link targets.
sortLabel	sortLabel field string to string	Maps a sort field code to its display label: "name" to "Name", "type" to "Type", "mtime" to "Modified", "size" to "Size", "date" to "Date". Unknown fields fall back to the raw field name capitalised. Empty string to "Modified" (the default).

Editing the templates — the basics

1. Find the templates dir. On this host it's `/etc/caddy/gallery-templates/`. The file is `gallery.tmpl`.
2. Edit the file in place. The module re-reads it on every request, so changes take effect immediately — no Caddy restart needed.
3. Test in a browser. The next request will use the new template. If the template has a parse error, the module will return 500 with the parser error in the response body. Fix the syntax and try again.
4. Revert to the bundled version. If you want to go back to what the binary ships with, delete the on-disk file: `sudo rm /etc/caddy/gallery-templates/gallery.tmpl`. The next request falls back to the bundled constant.

Walkthrough: change the theme to dark mode

Since the CSS is now inlined in the same file as the HTML, you edit the `<style>` block directly in `gallery.tmpl` — no second file to coordinate, no sub-template indirection.

1. Open `/etc/caddy/gallery-templates/gallery.tmpl` in your editor.
2. Find the `<style>` block. It's the big CSS block near the top of the file, just after `<title>` `>...</title>`. Search for `html, body { background: #f3f6f7;` to find the body color.

3. Change the colors inline. The CSS is plain CSS — no preprocessing. Most of the theme colors are concentrated in the first 50-100 lines.
4. Save the file. The change takes effect on the next request (no Caddy restart needed — the module re-reads the on-disk template on every `loadTemplate` call).
5. Hard-reload in the browser to bypass the HTML cache.

A common set of swaps for dark mode (find/replace these in the `<style>` block):

Light (default)	Dark variant
<code>html, body { background: #f3f6f7; }</code>	<code>html, body { background: #1a1a1a; color: #ddd; }</code>
<code>main { background: white; }</code>	<code>main { background: #222; }</code>
<code>header { border-bottom: 1px solid #e5e9ea; }</code>	<code>header { border-bottom: 1px solid #333; }</code>
<code>.chip { background: #f3f6f7; border: 1px solid #e5e9ea; }</code>	<code>.chip { background: #2a2a2a; border: 1px solid #3a3a3a; }</code>
<code>a { color: #006ed3; }</code>	<code>a { color: #4ea3ff; }</code>

Walkthrough: add a “Created” column to the image tiles

If you want to show the file’s created time alongside the modified date (the template currently shows Date which is the modified date):

1. Note: `FileView.Date` is the only date field exposed. The `FileInfo` struct in the scanner *does* have `ModTime` (in nanoseconds) but no `CreatedTime`. Adding a “Created” column would require a code change — extend `FileView` with a `Created` string field, populate it in `RenderPage` from `info.ModTime` (or from `os.Stat birthtime` via `syscall.Statx` on Linux), then reference it in the template as `{{.Created}}`.

In other words: this is a code change, not a template-only change. The template is fully driven by the Go struct fields — anything you want to display has to be in the struct.

Where the bundled templates are defined in source

If you want to read the source (or fork and customise):

File	Constant	Lines (approx)
<code>render.go</code>	<code>galleryTemplate</code>	line 392, ~574 lines (HTML + inlined CSS + inlined JS)

A single constant. CSS and JS are inlined inside `<style>` and `<script>` blocks respectively. To customise: edit the constant, rebuild the module (`./build.sh`), restart Caddy (`sudo systemctl restart caddy`). The on-disk template is written from the new constant on the next startup.

Upgrading from a pre-inlining install (Phase 16 to Phase 17)

If your site was running the old 3-file template split (`gallery.tmpl` + `style.css` + `lightbox.js`) and you upgraded to the new inlining build, the on-disk files are in an inconsistent state:

- `style.css` and `lightbox.js` from the old build are now dead weight. `writeBundledTemplates` removes them automatically on the next Provision after upgrade. Safe to ignore.

- The on-disk `gallery.tmpl` from the old build still has the old `{{template "lightbox.js" .}}` references, which no longer work. The new `loadTemplate` will fail to parse this file and the gallery will 500.

One-time fix on upgrade:

```
1 sudo rm /etc/caddy/gallery-templates/gallery.tmpl
2 sudo systemctl restart caddy
```

The next Provision writes the new inlined template. After that, the on-disk file is the canonical inlined version, and any operator edits to it become live overrides.

Upgrading the bundled template content

`writeBundledTemplates()` deliberately does NOT overwrite an existing `gallery.tmpl` on disk — that’s the operator-override contract. When the bundled template’s contents change (e.g. new CSS rule, new template branch, fixed layout bug), the on-disk file is NOT updated automatically.

To pick up the new bundled content:

```
1 # 1. Save any local customisations (if you have any)
2 diff /etc/caddy/gallery-templates/gallery.tmpl /home/osmanj/projects/
   caddy_image_gallery/render.go
3 # 2. Delete the on-disk file
4 sudo rm /etc/caddy/gallery-templates/gallery.tmpl
5 # 3. Restart Caddy so the next Provision runs writeBundledTemplates
6 sudo systemctl restart caddy
7 # 4. (Optional) Re-apply your local customisations
```

This workflow is intentional: operators who have customised the template (e.g. themed dark mode) keep their changes across upgrades. Operators who haven’t customised just get the “fresh bundled version” after the `rm` + `restart`.

Future enhancement (not v1): `writeBundledTemplates` could write a content hash into a sidecar file, and on Provision, if the sidecar hash doesn’t match the bundled hash, overwrite the on-disk file. This would auto-update without breaking the operator-override contract (the operator could still delete the on-disk file to fall back to the bundled version, OR write a sidecar with a custom hash to pin to their version).

Troubleshooting

Edit took effect but the page looks the same. Hard-reload (`Cmd-Shift-R` / `Ctrl-F5`). The browser may have cached the previous HTML.

Edit took effect but the page is a 500. Your template has a parse error. The response body will contain the Go `html/template` parser error. Common causes: - Unclosed `{{if}}` / `{{range}}` / `{{with}}` blocks - `{{end}}` mismatch (most common — Go templates require `{{end}}` to close every block) - Calling a method that doesn’t exist on the data type (e.g. `{{.Foo.Bar}}` when `Foo` is a string) - An unescaped backtick inside a Go template comparison (backticks terminate the Go raw string the constant is in)

Edit took effect but layout is broken. You removed or re-ordered a structural element. The CSS selectors and the JS `querySelector` calls expect a specific DOM shape — search for the class name in the template to see what’s expected.

Want to test a new template before deploying it. Stage the edit in a new file at, say, `/etc/caddy/gallery-templates/gallery.tpl.staging`, then copy it over the live file once you're happy. Or `curl` the gallery to see the live HTML and check it with a browser inspector before saving.

Got a 500 immediately after upgrading. You have the pre-inlining on-disk `gallery.tpl` from a previous build. See the “Upgrading from a pre-inlining install” section above.

Dark mode + theme toggle

The template supports three themes:

1. Auto (default) — follows the visitor's OS preference via the `prefers-color-scheme` CSS media query. macOS, Windows, iOS, and Android all expose this preference; if the visitor's OS is in dark mode, the gallery renders in dark mode automatically.
2. Light — forces light mode regardless of OS preference.
3. Dark — forces dark mode regardless of OS preference.

The choice is presented as a small toggle button group in the page header. Three buttons: a gear icon (Auto), a sun icon (Light), a moon icon (Dark). The current state is marked with `aria-pressed="true"` and a slightly inset background.

Position (Phase 50/51): the toggle was originally at the top-LEFT of the header (just to the left of the page title). As of Phase 50, it sits at the top-RIGHT, where the old sort-order indicator used to be. The sort indicator itself was removed in Phase 50 (the sort UI still exists as the sort-bar below the header — the active sort field is still visible there).

Mobile layout (Phase 50/51): on screens $\square$ 600px wide, the header stacks vertically via `@media (max-width: 600px) { .header-top { flex-direction: column; }`. The visual order on mobile is:

1. Theme toggle (top, right-aligned via `align-self: flex-end`)
2. h1 + meta line (below, full width)

This is achieved with CSS `order` on the flex children, not by reordering the HTML source (the source order stays `h1` $\square$ meta $\square$ toggle, which is better for screen readers):

```
1 .header-main { order: 2; }
2 .theme-toggle { order: 1; align-self: flex-end; }
```

The toggle's row order (`order: 1`) puts it above the header content (`order: 2`) on mobile; the `align-self: flex-end` keeps it right-aligned. On desktop (the default), neither order nor `flex-direction` applies, so the layout reverts to the normal horizontal flex with the toggle at the right (via the `justify-content: space-between` on `.header-top`).

The choice persists across visits via `localStorage` under the key `gallery-theme`. Values stored: `auto` | `light` | `dark`. `auto` is the default (no attribute set on `<html>`); `light` and `dark` set the `data-theme` attribute on `<html>`.

No flash of wrong theme

There's a tiny inline `<script>` in the `<head>` that reads `localStorage` and applies the `data-theme` attribute to `<html>` BEFORE the body paints. This runs synchronously during HTML parsing, so the CSS already sees the right theme when the body renders. No flash of light theme when the visitor has chosen dark.

How dark mode is implemented

All colors are defined as CSS custom properties (CSS variables) in the `:root` selector at the top of the template's `<style>` block. There are ~16 tokens, divided into semantic groups:

Group	Tokens
Backgrounds	<code>--bg</code> (page), <code>--bg-card</code> (cards), <code>--bg-chip</code> (chips), <code>--bg-hover</code> (chip hover), <code>--bg-active</code> (active chip)
Text	<code>--fg</code> (primary), <code>--fg-muted</code> (secondary), <code>--fg-faint</code> (tertiary), <code>--fg-disabled</code> (disabled)
Borders & shadows	<code>--border</code> , <code>--border-strong</code> , <code>--shadow</code> , <code>--shadow-strong</code>
Accents	<code>--accent</code> (links + borders), <code>--accent-hover</code> , <code>--accent-bg</code> (button fills — separate from <code>--accent</code> so the dark-mode button can be a muted darker blue without dimming link text)

The dark mode override is just a second block of token assignments that applies when:

- `@media (prefers-color-scheme: dark) { :root:not([data-theme="light"]) { ... } }` — the OS is in dark mode AND the user hasn't explicitly picked light. The `:not([data-theme="light"])` selector is what makes the “Auto” state work: when the OS is dark but the user picked Light, the dark tokens don't apply.
- `[data-theme="dark"] { ... }` — manual dark mode regardless of OS preference. Triggered by clicking the moon icon; choice persists in `localStorage`.

Why `--accent` and `--accent-bg` are separate tokens: the accent color serves two visual roles — text/borders (good as a bright color on a dark bg, e.g. `#4dabff`) and button fills (looks glaring as a bright color, should be muted, e.g. `#3b6fb6`). Splitting them lets each role have its own dark-mode value.

Dark mode refinements (Phase 40–42)

The first dark-mode pass had two issues:

- Chip bg too light (Phase 40): the chips (directories, other files, sort buttons, page buttons) had a `--bg-chip` value (`#1d1d1d` in dark mode) that was visibly lighter than the page bg (`#1a1a1a`), making the chips stand out as bright blobs. Fixed by setting `--bg-chip: #1a1a1a` (same as `--bg`) in dark mode, so chips blend in with the page; only the border + text show. This mirrors the light-mode behavior (where `--bg-chip` already equals `--bg`).
- Hardcoded colors in 11 CSS rules (Phase 41): a regex-based refactor (the original dark-mode implementation) caught 18 rules but missed 11 more that used hardcoded light-mode hex values. After a comprehensive sweep, all CSS rules now use `var()` token references. The only `#hex` colors left in the on-disk template are: the token definitions themselves (light + dark overrides), the video tile placeholder gradient (theme-independent), and the play button colors (white-on-dark, theme-independent).

Customizing colors

To override a color, edit the on-disk template at `/etc/caddy/gallery-templates/gallery.tmpl` and change the token value in the `:root` block. For example, to make the accent color green instead of blue:

```

1 :root {
2   --accent: #2e8b57;      /* was #006ed3 */
3   --accent-hover: #3aa86a; /* was #0095e4 */
4   --accent-bg: #2e8b57;   /* was #006ed3 — used by active sort/page buttons */
5 }

```

To add a new dark-mode-specific color, define it in all three blocks (:root, the media query, and [data-theme="dark"]).

Lightbox (always dark)

The lightbox overlay (the full-screen image/video viewer) has its own dark colors that are theme-independent — the dark background and white controls work in both modes. This is intentional: a dark overlay focuses attention on the content regardless of the page theme.

Header meta line format

The page header (below the page title) shows a meta line summarizing the directory’s contents. The format (Phase 44/45) is:

1	34 images · 8 videos · 2 other files // (8.3 MB total) // · 26 directories · 50 per					
2	page					
3	count	count	count	total size	count	page
4	size					
5	(when			(ALL files:		
6	TotalPages			images + videos		
7	> 1: also			+ other files)		
	Page X of Y")					"

Meta items, in order:

1. Image count — N images. Always shown.
2. Video count — · N videos. Shown only when N > 0.
3. Other files count — · N other files. Shown only when N > 0.
4. Total size — // (X.X KB total) //. Wrapped in // separators (visually distinct from the · separator used for the file counts). Represents the SUM of Size for all files in the directory: images + videos + other files. Excludes subdirectories. The literal word total follows the size inside the parens, making the meaning clear. Pre-formatted via the humanSize() helper (B / KB / MB / GB).
5. Directory count — · N directories. Shown when the user is in a subdir (N = subdirs of the current dir) or when there are subdirs in the root listing.
6. Per page size — · 50 per page. Always shown.
7. Page indicator — · Page X of Y. Shown only when totalPages > 1 (multi-page gallery).

Why the size uses // separators instead of ·: the // visually distinguishes the size from the other meta items (which all use ·). The size is conceptually different — it’s a quantity (bytes), not a count. The // is a “this is special” marker.

Implementation: the meta line is rendered by the template at the top of the <style> block’s wrapping <header>. The size segment uses three separate elements (one for each //,

one for the parens) so the browser's flex gap: 0.5rem adds visual spacing between them.

Open-in-new-tab button

The template has a small  button on each image/video tile (top-right corner of the card) that opens the file's URL in a new browser tab. Added in Phase 47; refined in Phase 52.

Visual: - Position: top: 6px, right: 6px inside the .card - Size: 28×28 px - Background: rgba(255, 255, 255, 0.85) — light translucent - Dark border: border: 2px solid #000 (Phase 47) so the button stands out more - Arrow color: #111111 (Phase 52) — a fixed dark color, NOT the theme-aware --fg token - The character itself is north east arrow (U+2197 NORTH EAST ARROW, displayed as the arrow glyph in the bundled template) - Default opacity: 0.5 (subtle). On hover/focus, opacity goes to 1.0 and the button scales up slightly.

Why the arrow color is fixed #111111 (not var(--fg)): the button has a light translucent background (rgba(255,255,255,0.85)) that stays light over ANY page background (light or dark). A dark arrow on a light button background is always visible — no need to adapt to the page's theme. The --fg token stays at its current values (#111111 light / #e5e5e5 dark) and continues to be used by other elements (hl, body, meta, sort buttons, etc.) — only .open-btn is excluded from the theme-aware color rule.

Behavior: - Click or Enter on the button  window.open(href, '_blank', 'noopener,noreferrer') opens the file in a new tab - The noopener,noreferrer flags are a security best practice (prevents the new tab from accessing window.opener) - The click event is stopPropagation'd so it doesn't also trigger the parent's "open in lightbox" handler

Lightbox controls

The lightbox overlay has 4 buttons + 2 text labels:

Element	Position	Behavior
.lb-close (×)	Top-right (inside .lb-controls pill)	Closes the lightbox; Esc key
.lb-open ()	Top-right (inside .lb-controls pill, left of close)	Opens the current image/video in a new tab
.lb-prev (◀)	Left edge, vertically centered	Previous image; Left-arrow key
.lb-next (▶)	Right edge, vertically centered	Next image; Right-arrow key
.lb-counter	Bottom-left (or bottom-center on mobile)	"N / total" text
.lb-caption	Bottom-center	The current file's name

The .lb-controls rounded pill (Phase 48): the open () and close (×) buttons are wrapped in a single flex container .lb-controls so they appear as a single rounded pill at the top-right of the lightbox. The container has: - Position: top: 1rem, right: 1.5rem - Background: rgba(255, 255, 255, 0.92) (light pill on the dark lightbox) - Border: 2px solid #000 - Border-radius: 10px (the "rounded box") - Padding: 4px around the buttons - Display: flex (lays out open + close side-by-side, perfectly aligned vertically and horizontally)

The two buttons inside have: - position: static (flex lays them out, not absolute) - Transparent background (the container provides it) - No individual border - 28×28 px - Dark icon color (#1a1a26) - border-radius: 6px on each button (rounded corners within the pill) - Hover: subtle dark background tint (rgba(0, 0, 0, 0.08))

Why a pill instead of separate buttons: a single rounded container gives the two related actions (close + open) a visual unity — they’re both “exit the lightbox” actions (close or view-on-its-own), and grouping them makes that clearer.

The prev/next buttons are NOT in the pill — they’re at the left/right edges of the lightbox, vertically centered. These are navigation actions (different from exit actions), so they get their own positioning.

Why a dark border: like the tile `.open-btn`, the pill has a black 2px border so it stands out clearly against the dark lightbox background. The light pill on a dark bg with a dark border is a strong visual “this is a control” signal.

Keyboard navigation: - Escape ☐ close - Left arrow ☐ previous image - Right arrow ☐ next image - (Clicking on the image itself also goes to next, like carousel UIs)

The `data-theme` attribute on `<html>` is NOT read by the lightbox CSS — the lightbox is intentionally theme-independent (dark always, with light controls). This is the same design choice as the lightbox bg and counter/caption: focus on the content, regardless of page theme.

Video poster (Phase 63): when a video tile has a generated thumbnail (Phase 62’s ffmpeg pipeline), the lightbox video element sets its HTML5 poster attribute to the thumb URL (extracted from the same `<img class="thumb-img">` element on the tile). The browser shows the poster image immediately when the video opens in the lightbox — the user sees the video’s first frame as a still image, then on click the video swaps to playback. This is the same mechanism YouTube uses to show a thumbnail before a video plays.

If `no_video_thumbs` is set OR `ffmpeg` is missing, the tile has no `<img class="thumb-img">` so the JS can’t find a poster URL — the poster attribute is simply not set, and the browser shows its default (black frame, or the first decoded frame if `preload="metadata"` is enabled).

The poster URL points at the same cached WebP that’s used for the tile thumbnail, so no extra server work or storage is required.

Building the PDF locally

The PDF book (`caddy-image-gallery-book.pdf`) is built from the markdown files in `docs/` via `pandoc + xelatex`. The full command is wrapped in a script at the project root: `build-docs.sh`.

To rebuild the PDF locally:

```
1 ./build-docs.sh
```

The script: - Works from any directory (uses `BASH_SOURCE` to find its own location and the project root) - Verifies prerequisites (`pandoc`, `xelatex`) before running - Verifies all source files exist before running - Runs `pandoc` with the right flags (`--pdf-engine=xelatex`, `--include-in-header=preamble.tex`, `--listings`) - Cleans up intermediate files (`*.aux`, `*.log`, etc.) after - Prints summary info (output path, file size, page count)

Prerequisites: - `pandoc` - `xelatex` (usually in the `texlive-xetex` package) - Install on Debian/Ubuntu: `sudo apt-get install pandoc texlive-xetex texlive-fonts-recommended` - Install on macOS: `brew install pandoc mactex`

Why a script instead of running `pandoc` directly: the `pandoc` command has several flags (`--include-in-header=preamble.tex` references the font config; `--listings` is needed for code block formatting; the source file order matters). The script encapsulates all of this so you don’t have to remember the incantation.

The font config (preamble.tex + docs/fonts/) references absolute paths for fontspec's Path = option. If you move this project to another machine, update the Path = line in preamble.tex to match the new location.

Sort & Pagination

The gallery's image grid respects three URL query parameters. All three are optional. They're a stable URL API — bookmarking a `?sort=size&order=desc&page=3` link works and gives the same view on every visit.

Query parameters

Param	Values	Default	What it does
?sort=	name / type / size / mtime	mtime	What field to sort images by. The URL also accepts <code>?sort=date</code> (treated as <code>?sort=mtime</code>) for back-compat. See “Aliases” below.
?order=	asc / desc	desc	Sort direction. desc for mtime is newest-first; desc for name/type/size is Z-A / largest-first.
?page=	integer >= 1	1	Which page of results to show. 1-based. Out-of-range or non-numeric values fall back to 1.

The URL parameter is preserved across sort changes: clicking “Name” on `?sort=mtime&order=desc&page=2` becomes `?sort=name&order=asc&page=2` (the toggle flips the order for the new field; the page stays the same).

The directory strip and other-files strip are NOT affected by the sort. They always render in:

- Directories: case-insensitive alphabetical (splitFiles re-sorts them on every render)
- Other files: scanner order (same default as the image list — newest-first by mtime)

This is intentional: the dirs strip is a navigation aid and should be stable, not reshuffle when the user changes the image sort. See Phase 15 in the plan for the reasoning.

Sort indicator in the header

The header shows the current sort + a reset link:

- On the default sort (mtime desc), the indicator is a plain `<span>` — clicking it would just reload the same page, so it's not a link. It reads Sort: Modified `$\downarrow$`.
- On any other sort, the indicator is a link to ? (no query params, which is the default). It reads e.g. Sort: Size `$\uparrow$`.

Sort buttons

The sort bar at the top of the header has four buttons: Name, Type, Modified, Size. Each shows a `$\uparrow` or `$\downarrow` arrow if it's the currently active sort. Clicking the active button toggles the order; clicking an inactive button switches to that field at the direction that's the “natural” opposite of the current order (so you don't get the same direction twice in a row).

The button labels are produced by the `sortLabel` template function (see the Templates doc for the full map). To add a new sort field:

1. Add a case to `sortLabel` in `render.go`
2. Add a case to the `sortFiles` function (the sort implementation)
3. Add a button to the `gallery.tmpl` template

It's a code change in three places; the template is just the visible button.

Pagination

50 images per page (constant `PageSize = 50` in `render.go`, not configurable via env var or Cad-file — it's a code constant).

Pagination links in the live template: - The “First” and “Prev” pagination buttons (rendered by the template with double- and single-left-arrow icons, when `{{.HasPrev}}` is true) - The “Next” and “Last” pagination buttons (single- and double-right-arrow icons, rendered when `{{.HasNext}}` is true) - Current page indicator in the middle

The page numbers in the URL are 1-based, not 0-based. `?page=0` falls back to `?page=1`.

Examples

```
1 /images/                                # default: mtime desc, page 1
2 /images/?sort=name&order=asc            # alphabetical
3 /images/?sort=size&order=desc          # largest first
4 /images/?page=3                        # third page of default sort
5 /images/?sort=type&order=asc&page=2
```

Aliases

`?sort=date` is an alias for `?sort=mtime` — both sort by the file's modification time. The two values produce identical results:

```
1 /images/?sort=date&order=desc          # = /images/?sort=mtime&order=desc
2 /images/?sort=date&order=asc           # = /images/?sort=mtime&order=asc
```

Why have the alias? The two names describe the same thing from different angles — “mtime” is the technical field name (modification time, from the filesystem), “date” is the user-friendly name (the date the file was last changed). The “Modified” sort button in the UI emits `?sort=mtime` (the technical name), but the URL API accepts both for back-compat with bookmarks and external links that might use either name.

Implementation: `parseSort` in `render.go` accepts “date” in its switch statement and treats it identically to “mtime”:

```
1 switch field {
```

```
2 case "name", "type", "date", "mtime", "size":  
3     // all valid; "date" is an alias for "mtime"  
4 default:  
5     field = "mtime" // unknown fields fall back to the default  
6 }
```

In `sortFiles`, both `"mtime"`, `"date"`, and `""` (the empty default) share the same sort branch — sort by `ModTime` per the `order` param. So functionally there's no distinction between them.

When to use which: if you're writing a new integration or bookmark, prefer `?sort=mtime` (it's the canonical name and what the UI button emits). `?sort=date` is fine too — just pick one and stick with it for consistency.

Stability guarantees

- The URL parameter API is stable. Bookmarking works. The four sort fields and two orders are part of the public contract; renaming or removing one is a breaking change.
- Adding a new sort field is non-breaking (just adds a new option).
- The default sort may change in a future major version, but the `?sort=mtime` URL will always be honored.
- Page size is a code constant and may change in future versions. URLs with `?page=N` are stable relative to the *current* page size — if the page size changes from 50 to 100, page 3 of the old URL might now be page 2 of the new layout, but no image is lost or duplicated (it's the same image set, just packed differently).